KCE PATHWAYS · GRADES 4–5 · UNIT 4: SHAPES AND SPACE · WEEK 17 · SCIENCE

Energy Resources: Renewable and Nonrenewable

Name: _____ **Date:** _____

PRE-LESSON WARM-UP · DO THIS BEFORE YOUR COMPUTER LESSON

Pre-Lesson Warm-Up: What Do I Already Know?

Answer with what you know right now. It is OK to guess!

1. **Review (Week 16):** A wave passes a toy boat floating in a pool. What does the wave carry across the water, and what happens to the boat?

2. Where do you think the electricity for the lights in your home comes from? Explain your guess.

3. List as many energy sources as you can, such as the Sun. Circle any that you think could run out someday.

4. Guess: How did coal, oil and natural gas form? What were they made from?

5. Write one thing your family could do to use less energy at home.

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POST-LESSON CHALLENGE · DO THIS AFTER YOUR COMPUTER LESSON

Post-Lesson Activity: Show What I Learned

Use evidence from today's lesson. Write in complete sentences.

1. Sort these resources into **renewable** or **nonrenewable**: wind, coal, uranium, sunlight, natural gas, wood that is replanted, oil, geothermal heat. Then explain how you decided.

2. **Two steps (EIA, 2025 data):** Natural gas made about 41% of U.S. utility-scale electricity and coal made about 17%. (a) What percent did they make together? (b) Is that more or less than half?

3. **Find the mistake:** "Oil comes from dinosaur bones, and nuclear energy is a fossil fuel." Fix BOTH mistakes, using evidence from the lesson.

4. Choose one energy resource. Write one **benefit** and one **cost** (a trade-off), and explain why no resource is perfect.

★ **Challenge:** Be an Energy Detective at home, with a grown-up. List two ways your family could save energy: one **conservation** action and one **efficiency** choice. Label each one.

Looking Ahead: Next week: **Earth's Systems.** You will explore the geosphere, hydrosphere, atmosphere and biosphere, and how they interact in a volcano, a hurricane and a forest. Bring this question: where is most of Earth's water?